



Regional Sales Analysis with DataCo

Business Intelligence

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1. Problem, user and decision

Selected commercial line from W1

Research question

How do the best-selling product categories vary across regions, and what factors are associated with lower sales volumes?

Proposed user

Commercial analyst

Decision

Which region–category combinations should be prioritized for further analysis before making assortment or promotional decisions?

Candidate A (sales) was retained over Candidate B (logistics) because it better fits the commercial variables and the intended decision.

The analysis is historical and descriptive: associations are not interpreted as causal effects.

2. Seven-step analytical framework

A direct mapping from the business decision to measurable evidence

1 • Decision

Prioritize **region–category combinations** for further commercial analysis

2 • Success

Identify **meaningful and comparable sales patterns** that can support commercial prioritization.

3 • Questions

Compare **sales volume, category share, units per order and discount patterns** across regions.

4 • Process

Order → item → status → shipment

5 • Entities

Customers, orders, items, states

6 • Representation

1 row = 1 order item

7 • Measurement

Define the KPIs used to compare regions and categories. **Main KPI: Units per Order**, complemented by Recorded Units and Category Share.

3. Data preparation and the Five Cs

DataCo Supply Chain • 180,519 order-item records • 53 variables

Observation unit

1 row = 1 Order Item
Order Item Id is unique.

Commercial subset

172,765 retained items
62,897 represented orders

Status exclusions

3,692 CANCELED
4,062 SUSPECTED_FRAUD

Clean

No duplicate item IDs; essential fields validated.

Consistent

Temporal coverage differs across regions/categories.

Conformed

No external join; category ID/name consistency checked.

Current

2015–2018: suitable for historical analysis, not 2026 operations.

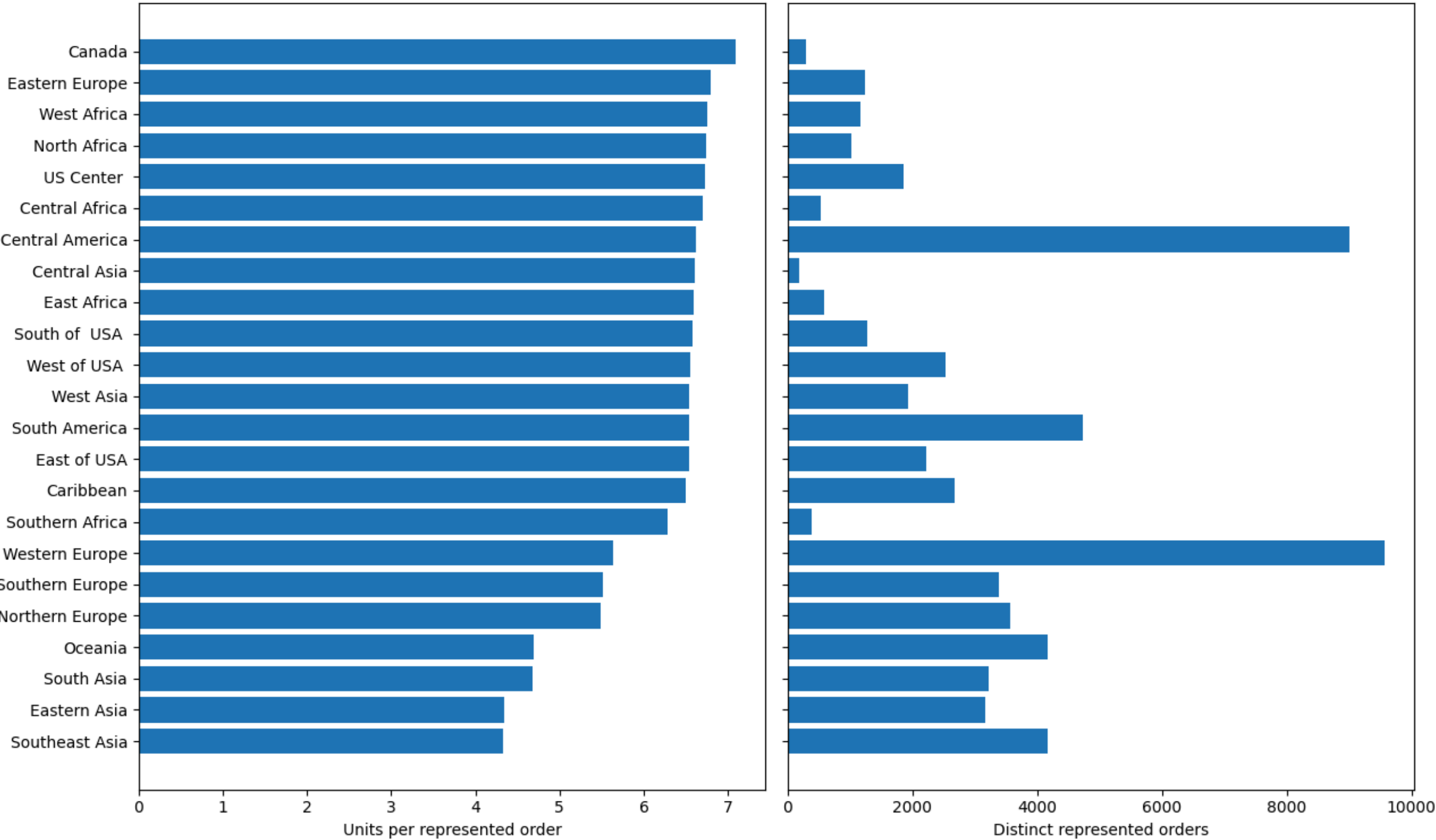
Comprehensive

No stock, unmet demand, campaign exposure or competitor data.

4. EDA – Regional volume is not regional performance

Raw volume and normalized purchasing intensity tell different stories

Purchase intensity and sample size — eligible records



Canada

2,063 units
7.09 units per order

Western Europe

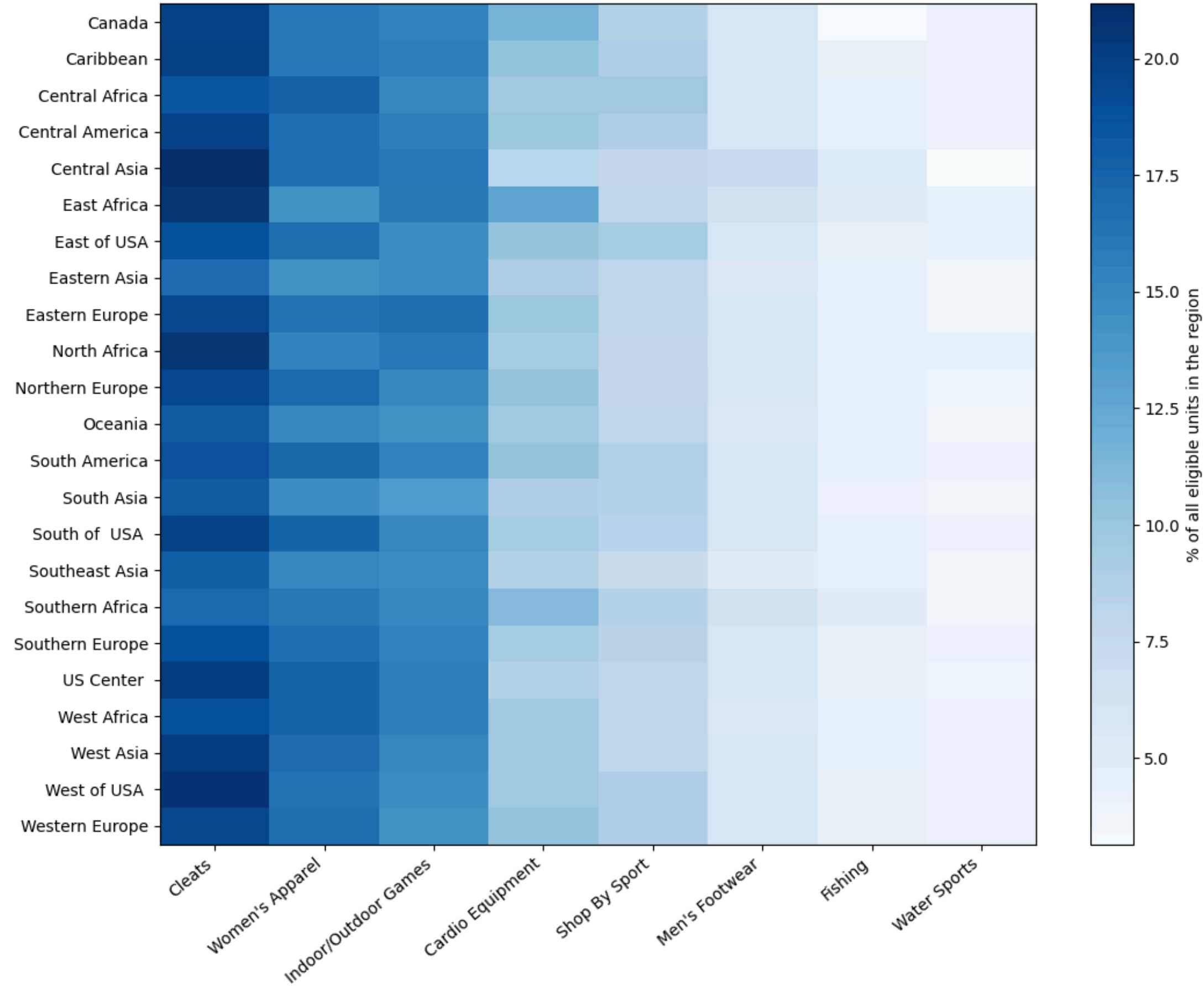
9563 orders
5.64 units per order

Central America

59,540 total units (volume)
(orders x units per order)

EDA visual — Category mix across regions

Recorded unit share within each region — top 8 categories overall



What the chart shows

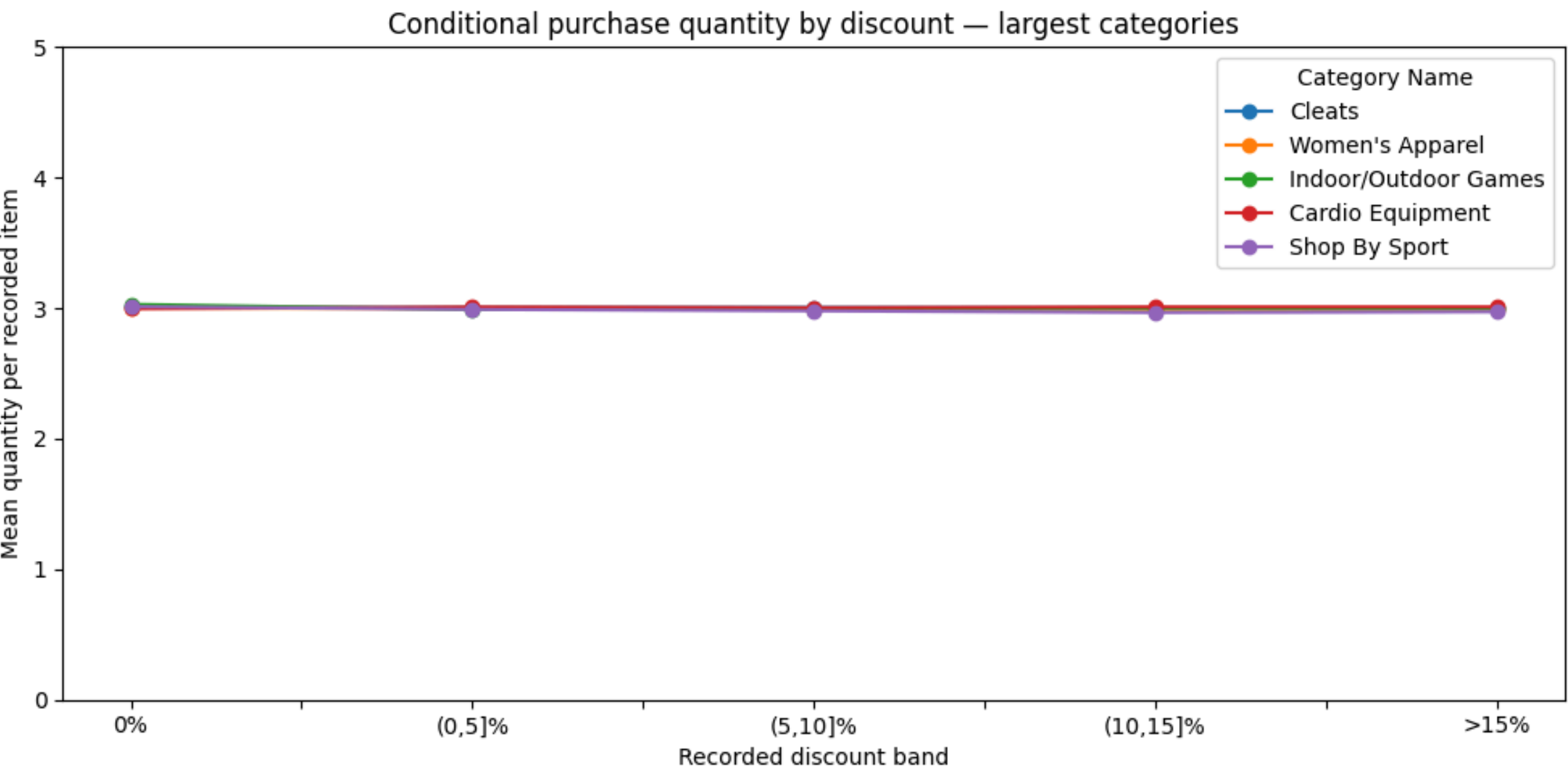
Category composition is relatively similar across many regions: Cleats, Women's Apparel and Indoor/Outdoor Games account for large shares of recorded units.

Interpretation boundary

These shares describe the observed DataCo transactions. They do not measure total regional demand or market potential.

EDA visual – Discount bands show little separation

Conditional purchase quantity remains close to three units per recorded item across the largest categories



Observed pattern

Across the five largest categories, the max-min difference in mean item quantity ranges from only 0.008 to 0.045 units.

Commercial implication

The descriptive evidence does not justify recommending deeper discounts. Causal effects are not identified.

5. Robustness checks and discount analysis

The main descriptive findings are stable, but they do not support causal recommendations

Status sensitivity

Largest absolute change in regional category share after excluding canceled/fraud items:
≈ 0.54 percentage points

Discount bands

Across the five largest categories, max-min mean item quantity differs by only:
0.008–0.045 units

Commercial implication

The observed discount pattern does not justify recommending deeper discounts.

- Monetary arithmetic checks found no discrepancies above 0.05 in the retained subset.
- Lower recorded sales are not proof of weak underlying demand.
- Alternative explanations include assortment, product availability, regional activity, order states and unequal observation windows.

6. Main limitation – unequal temporal coverage

Comparability is the central constraint on the regional interpretation

Material finding

Canada is observed only from Aug 2016 to Jan 2017, while other regions have different observation windows.

Why it matters

Units per order reduces scale differences, but it does not remove time or assortment confounding.

Required next check

Restrict future comparisons to overlapping periods and available categories, and report the restricted sample.

- Use volume, category share, units per order and sample size jointly.
- Review reporting coverage and assortment before making investment decisions.
- Seek inventory, campaign exposure/cost, margins and customer opportunity data before prescribing action.

7. Project continuity – ML and dashboard

Future ML task

A future model could forecast next-month recorded units for each region–category combination, although limited temporal coverage must be considered.

Dashboard proposal

The dashboard would help commercial analysts compare units, orders, units per order, and category share using filters such as month, region, and category.

Decision boundary

The results should support further investigation rather than automatic commercial decisions, and clustering could be used if forecasting is not feasible.

8. Conclusions

DataCo supports a descriptive comparison of regional sales patterns, with Western Europe leading in recorded volume and Canada showing higher units per order on a smaller sample.

The analysis did not provide enough evidence to reach a definitive conclusion about which regions or categories perform better, mainly because temporal coverage and assortment differ across the dataset.